

de Rham decomp Thm.

Recall. $\text{Hol}(x) = \{ P_{01}^\sigma : \sigma \text{ loop @ } x \}$.

Def. M (ir) reducible if $\text{Hol}(x) \curvearrowright T_x M$

Thm. (de Rham '52) if M conn + complete + $\pi_1(M) = 0$ Riem mfd,
 then $\wedge$ $M \cong_{\text{isometry}} M_0 \times \dots \times M_k$ for M_0 euclidean (possibly n pt)
uniquely w/ product metric $\hookrightarrow$ not irr. so would split off into 1-dim
 and M_i so all simply conn + complete + irr. Riem mfd's.

Cor. S^{n+1} irreducible holonomy rep.

Non/ex. $(\text{circle}) \subset \mathbb{R}^n$ not complete. flat

Non/ex. flat torus $\mathbb{R}^2/\Lambda$ for $\Lambda = \mathbb{Z}(1,0) \oplus \mathbb{Z}(x,y)$
 w/ $x \in (0, 1/2]$ and $x^2 + y^2 \geq 1$ $\nmid$
 is NOT a product ($\pi_1 \neq 0$).

Thm. (Berger '55) All holonomy irr simp conn mfd's have hol in $\text{Hol}(g)$
 dim M n $2n$ $2n$ $4n$ $4n$ 7 8 $2n$
 $\text{Hol}(g)$ $SO(n)$ $SU(n)$ $SU(n)$ $Sp(n) \cdot Sp(1)$ $Sp(n)$ G_2 $Spin(7)$

Local Picture

Rate. say $0 \neq T'_x \subseteq T_x M$ is invariant by $\text{Hol}(x)$.

put $T'_y = P^\sigma(T'_x)$ for $\sigma: x \rightarrow y$.

T'_y shows ind. of σ .

$$T' \stackrel{\text{def}}{=} \bigsqcup_{y \in M} T'_y.$$

Prop. T' diff

Pf. $y \in M$. (xi) normal coords @ y .

(ei) o-n $T_y M$ w/ $e_1, \dots, e_k$ or T'_y .

put $(X_i)_{\exp(xie_j)} = P_{\exp(xie_j)}(e_i)$ local frame.

$X_1, \dots, X_k$ sp local frame for T' .

(3)

Prop. $T^\#$ involutive.

Pf. $X, Y \in T(T')$. $[X, Y] = \nabla_X Y - \nabla_Y X \in T(T')$.
 if $\gamma(0) = p, \dot{\gamma}(0) = X_p$. $(\nabla_X Y)_p = \lim_{h \rightarrow 0} \frac{P_{h,0}^T(Y(\gamma(h))) - Y_p}{h}$. $\square$

Cor. By Frobenius, T' is completely integrable, i.e. get $M'(z)$ ^{for $\gamma \in T'$}
 maximal integral mfd (its tangent space is T') ~~for $\gamma \in T'$~~ (foliation).

Rank. From our normal coord picture, its totally geodesic.
 so M complete $\Rightarrow M'(z)$ is complete.

put $T'' = (T')^\perp$. T'' also invol under $\text{Hol}(x)$ since $\text{Hol}(x) \in O(T_x M)$.

(4)

Prop. $y \in M$. say M', M'' max int. mfd for $T', T'' @ y$.

$\Rightarrow \exists$ open $V = V' \times V''$ so V' open in M', V'' open in M''
 s.t. $g = g' \oplus g''$.

Pf. T' invol $\Rightarrow y^1, \dots, y^k, x^{k+1}, \dots, x^n$ ($x^j = c^j$ defn M').
 T'' invol $\Rightarrow x^1, \dots, x^k, z^{k+1}, \dots, z^n$ ($x^i = c^i$ defn M'').

$(x^1, \dots, x^n)$ forms a coord system since

$$\begin{aligned} M &\xrightarrow{\phi} \mathbb{R}^k \times \mathbb{R}^{n-k} \quad \phi = (x^i). \\ T' \oplus T'' &\xrightarrow{d\phi} \mathbb{R}^k \times \mathbb{R}^{n-k} \quad d\phi = (\pi', \pi''). \\ \ker \pi'(v) &= T'(v) \text{ so inj.} \end{aligned}$$

let $V = \{x: |x^i| < c\}_{i=1}^n$ for c small enough that (x^i) is homeo. (5)

let $V' = V \cap \{x^1 = \dots = x^k = 0\}$, $V'' = V \cap \{x^{k+1} = \dots = x^n = 0\}$.

WTS $g(x^1, \dots, x^n) = \begin{pmatrix} g'(x^1, \dots, x^k) \\ g''(x^{k+1}, \dots, x^n) \end{pmatrix}$. ~~WTS~~
 $m \leq k < n \Rightarrow \in T'$ and T'' so 0.

$T' \perp T''$ so $\begin{pmatrix} 0 & 0 \end{pmatrix}$. coords $\Rightarrow [X^m, X^n] = 0 \Rightarrow \nabla_{X^m} X^n = \nabla_{X^n} X^m$.
 $X^{l \leq k} \langle X^{i \leq k}, X^{j \leq k} \rangle = \langle \nabla_e X^i, X^j \rangle + \langle X^i, \nabla_e X^j \rangle = 0 + 0 = 0$.

Prop. T', T'' . M simply conn. $\mathbb{F}(x) = \mathbb{F}'(x) \oplus \mathbb{F}''(x)$ ^{normal subgroups}
 so $\mathbb{F}'(x)$ ($\mathbb{F}''(x)$) trivial on T_x'' (T_x'). (6)

Pf. $a \in \mathbb{F}(x)$. (ei) for $T_x M$ $\begin{matrix} e_1 \dots e_k & \text{sp } T' \\ e_{k+1} \dots e_n & \text{sp } T'' \end{matrix}$

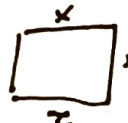
$$a = \begin{pmatrix} a_1 & \\ & a_2 \end{pmatrix}. \quad \text{let } a' = \begin{pmatrix} a_1 & \\ & 1 \end{pmatrix} \quad a'' = \begin{pmatrix} 1 & \\ & a_2 \end{pmatrix}.$$

WTS $a', a'' \in \mathbb{F}(x)$.

small lasso.

say τ induces a . if $\tau = \mu^{-1} \cdot \sigma \cdot \mu$ for $\mu: x \rightarrow y$
 set $\sigma' = \pi' \circ \sigma$ ($\sigma'' = \dots$) still loop @ y .
 set $\tau' = \mu^{-1} \cdot \sigma' \cdot \mu$ ($\tau'' = \dots$). τ' trivial on T_x'' .
 $P^{\tau' \cdot \tau''} = P^\tau$. so $\tau^{(1)}$ induces $a^{(1)}$.

Lem. M simply conn. parallel transport along τ is the product of parallel transport along a fin # of small lassos. (7)

Pf. $\tau \xrightarrow{H} \text{pt} = x$  $\xrightarrow{H} M$.

break into m^2 so $H(\square \text{ small})$ is in some $V' \times V''$.

set $\lambda(i, j) =$  $\text{ml } \tau(i, j) = H(\lambda(i, j))$.

 $\tau(m, m) \cdot \tau(m-1, m) \cdot \dots \cdot \tau(1, m) \cdot \tau(m, m-1) \cdot \dots \cdot \tau(1, 1)$
 parallel transport is just τ .

$\mathbb{F}'(x) = \{ a^{(1)} : a \in \mathbb{F}(x) \}$ finishes. (obv normal form matrix prod).

Def. $T_x^{(0)} \equiv \{v \in T_x M : \Phi(x) \cdot v = v\}$ Global Picture (8)

$$T'_x \stackrel{\text{def}}{=} (T_x^{(0)})^\perp \text{ invn by } \Phi(x).$$

$$T'_x = \bigoplus_i T_x^{(i)} \text{ pairwise orthog, invn, irr. subspaces.}$$

$$T_x M = \bigoplus_0^k T_x^{(i)} \text{ "canon / de Rham decomp"}$$

$$T^{(i)} \text{ invol dist for } T^{(j)}.$$

Prop. (1) $\exists V = V_0 \times \dots \times V_k$ for V_i open in $M_i(y)$. $\tau_v = \bigoplus \tau_{v_i}$. (1.1)

(2) M_0 locally euclidean. ($v_0 = \Pi$ 1-dim subsp. $g_{11} dx^1 dx^1$ norm assoc.)

(3) M simply conn. $\Rightarrow$ de Rham decomp unique up to order.
and $\Phi(x) = \prod \Phi_i(x)$ w/ $\Phi_0(x) = \text{identity}$
and Φ_i trivial on $T^{j \neq i}$, irr. on T^i .

Thm. M simply conn + conn + complete Riem mfd.

M reducible $\Rightarrow T', T''$.

fix θ . get $M' = M'(\theta)$, $M'' = M''(\theta)$.

we'll show $M \cong_{\text{iso}} M' \times M''$.

$\rightarrow$ complete + good subsp.

(9)

Pf. z_t curve in M $0 \leq t \leq 1$. $z_0 = \theta$.

project $\dot{z}_t$ to $T' \rightarrow \dot{z}_t'$.

put $X_t = P_{t \rightarrow 0}^{\dot{z}_t}(\dot{z}_t')$.

put x_t in M' w/ $x_0 = \theta$ and $P_{t \rightarrow 0}^{x_t}(\dot{x}_t) = X_t$.

Claim. X_t only depends on z_t if M simply conn.

if T , get $p': M \rightarrow M'$.

~~isom~~ $p = (p', p'')$

obv isometry
parallel transport.

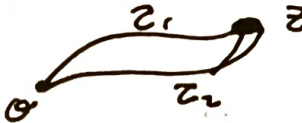
~~parallel transport~~ $\dot{z} = X_t p'$ and $\dot{z} = X_t p''$

Pf. (ctd). p local isometry since $\tilde{z} = \overset{\uparrow}{T}_\sigma M \overset{\uparrow}{T}'_\sigma X + \overset{\uparrow}{T}''_\sigma Y$. (10)
 and $p'(\tilde{\sigma})$ will be length X , as will $p'(Y)$.

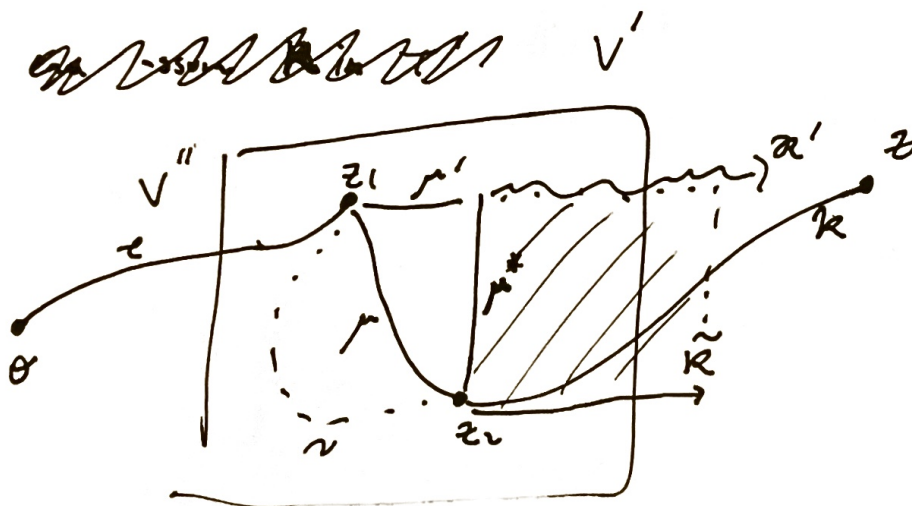
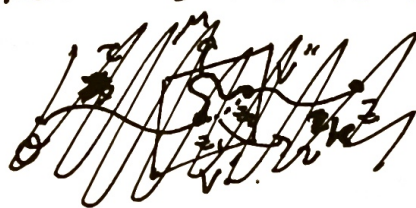
thus Riem covering map. also, $\pi_1(M') = 0$ since

if $\tilde{\sigma} \xrightarrow{H} \sigma$, then $p' \circ H$ is homotopy in M' .
 $\downarrow$
 $\text{in } M'$

so $M'_0 \times M''$ simply conn. so M universal cover $\cong M' \times M''$.

Pf. (claim). we show that if  in M , $z_1 \sim z_2$.
 then proj to $M'(\sigma) = M'$ lie the same end pt. (11)

but first note that z_2 can be obtained by a finite # of small steps, i.e.



⑥ break $\tilde{z}$ into z_i arcs fitting into $V'_i \times V''_i$.
 repeat ...

① $K \rightarrow \tilde{K} \subset M'(z_2)$.
 w/o closed end pt

② $\mu = (\mu', \mu'')$. μ' is
 proj of μ to $M'(z_1)$

③ $\mu'' \tau''$ - geodesic from z_2
 to end pt of μ' .
 (can't be arches)

④ get homotopy and K'

⑤ proj of $K \cdot \mu$ in $M'(z_1)$
 is $K' \cdot \mu'$.

proj. of $K \cdot z$ in $M'(z_1)$
 is $K' \cdot v'$.